

Comprehensive Literature Review: SBL Acoustic Positioning Systems, Configuration Management, and USV-AUV Collaborative Operations

1. Introduction and Scope

This literature review synthesizes current research relevant to the development of adaptive short-baseline (SBL) acoustic positioning management frameworks for collaborative unmanned surface vehicle (USV) and autonomous underwater vehicle (AUV) operations. The review covers six interconnected research domains:

1. Short-baseline acoustic positioning systems and configuration optimization
 2. Fisher Information Matrix (FIM) and performance prediction
 3. Observability analysis and array design
 4. Multi-sensor fusion and state estimation
 5. Energy management and optimization
 6. Collaborative USV-AUV systems and mission planning
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2. SBL Acoustic Positioning Systems: Fundamentals and Applications

2.1 System Principles and Architecture

Foundational Work: Wikström et al. established the conceptual framework distinguishing acoustic positioning systems (SBL, USBL, LBL) by baseline length and transducer configuration. SBL systems employ multiple transducers with spacing on the order of tens of meters, providing intermediate accuracy performance with the advantage of moving baseline compatibility.[web:1]

Key Characteristics:

- Three or more transducers measure time-of-flight from submerged transponder
- Position calculated via trilateration
- Accuracy improves with increased baseline spacing (unlike USBL)
- No seafloor infrastructure required (advantage over LBL)
- Suitable for mobile platforms (USVs)

Performance Metrics: Typical SBL accuracy: 1-3 meters depending on baseline length and environmental conditions.[web:45] Extended baseline enables accuracy approaching LBL systems while maintaining vessel mobility.

2.2 Operational Advantages and Constraints

Advantages:

- Real-time positioning capability
- No permanent infrastructure deployment
- Scalable accuracy with baseline adjustment

- Compatible with dynamic operations

Limitations:

- Acoustic noise from water column, biological sources, and surface interactions
- Multipath propagation: signals reflecting off water surface and seafloor create arrival time ambiguities
- Sound speed variability ($\pm 2\text{-}3\%$ with salinity and temperature changes)
- Power consumption for continuous modem operation
- Limited data bandwidth (250-4,500 bit/s for typical modems)
- Hardware failures in transducers or electronics

3. Fisher Information Matrix Analysis for Positioning Systems

3.1 Theoretical Foundation

Seminal Contributions: The Fisher Information Matrix (FIM) provides fundamental bounds on estimation accuracy through the Cramer-Rao Lower Bound (CRLB). For positioning systems, the FIM encapsulates how measurement geometry and sensor properties determine achievable accuracy.

Mathematical Framework: For measurements $\mathbf{z}$ and unknown position parameters $\boldsymbol{\theta}$, the FIM is defined as:

$$\mathbf{F}(\boldsymbol{\theta}) = \mathbb{E} \left[-\frac{\partial^2 \ln p(\mathbf{z}|\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^T} \right]$$

The CRLB provides the lower bound on unbiased estimator variance: $\text{Var}[\hat{\boldsymbol{\theta}}] \geq [\mathbf{F}(\boldsymbol{\theta})]^{-1}$

3.2 Application to Underwater Acoustic Positioning

FIM for Range-Based Positioning: The research by Arrichiello et al. [web:2] demonstrates practical FIM calculation for underwater USBL systems. For a set of range measurements to modems at known positions, the Jacobian matrix relates changes in position to changes in measurements:

$$\mathbf{H} = \begin{bmatrix} \frac{x-x_1}{r_1} & \frac{y-y_1}{r_1} & \frac{z-z_1}{r_1} & \frac{x-x_2}{r_2} & \frac{y-y_2}{r_2} & \frac{z-z_2}{r_2} & \vdots & \vdots & \vdots \end{bmatrix}$$

The FIM for Gaussian measurement noise becomes: $\mathbf{F} = \mathbf{H}^T \mathbf{R}^{-1} \mathbf{H}$

where $\mathbf{R}$ is the measurement noise covariance matrix.

Key Insight: Determinant of FIM provides scalar measure of total information content. Larger determinant indicates better position estimate accuracy, lower determinant suggests ambiguous geometry.

3.3 Array Geometry Optimization via FIM

Recent Advances: Chen et al. [web:2] developed methodology for optimizing USBL array configurations for AUV formation positioning. The optimization approach:

1. Parameterizes array geometry (transducer positions)
2. Calculates FIM for candidate configurations
3. Optimizes trajectory of USV to maximize FIM determinant
4. Validates optimal trajectory in simulation and field tests

Practical Result: FIM-based optimization identified optimal orthogonal array configurations for AUV formation localization, with performance verified through Dubins path planning for the USV.

Extension to SBL: The methodology directly extends to SBL systems, enabling assessment of which transducer subset (2, 3, or 4 active) maintains sufficient information content for mission requirements.

3.4 Configuration Assessment Metrics

Determinant-Based Assessment:

- FIM determinant directly relates to position estimate covariance
- Lower determinant = larger position uncertainty region
- Useful for comparing different modem combinations

Eigenvalue Analysis:

- Eigenvalues of FIM indicate information content in different directions
- Small eigenvalues suggest poor observability in those directions
- Condition number (ratio of largest to smallest eigenvalue) indicates numerical conditioning

Practical Application: Lookup tables can be precomputed mapping:

- Modem configuration (subset of 4) → FIM determinant
- AUV position relative to USV → GDOP metric
- Mission accuracy requirement → minimum required configuration

4. Geometric Dilution of Precision (GDOP) Analysis

4.1 GDOP Theory and Underwater Applications

Foundational Theory: Breyer [web:60] established GDOP mathematical framework for triangulation-based positioning. GDOP represents how sensor measurement errors magnify into position estimation errors based on geometric configuration.

Definition: GDOP = scalar multiplier transforming measurement error into position error. Derived from Jacobian matrix condition:

$$\text{GDOP} = \sqrt{\text{trace}((H^T H)^{-1})}$$

Interpretation:

- GDOP = 1-2: Excellent geometry, small error magnification
- GDOP = 5-10: Good geometry, moderate error magnification
- GDOP > 20: Poor geometry, significant error magnification
- GDOP → ∞: Singular configuration, geometry provides no information

4.2 Singular Configurations in Underwater Systems

Coplanarity Constraint: A fundamental challenge in underwater positioning is that all transducers must be mounted on the surface, creating inherent coplanarity. This constraint prevents achievement of theoretical minimum GDOP.

Cone Configurations: Lu et al. [web:11] analyzed GDOP in single-cone configurations (transducers arranged as base of cone, target somewhere on cone surface). Result: GDOP becomes infinite when target lies on cone axis, creating positioning ambiguity.

Solution: Combined Cone Configuration: Proposed approach combines multiple single-cone configurations, resolving singularities through observation diversity. Key innovations:

1. Multiple cone configurations added together
2. Rotation applied to cones for different viewing angles
3. Global GDOP optimization over target trajectory (not just single points)
4. Demonstrated GDOP improvements through combined configuration

4.3 GDOP Optimization for SBL Systems

Dynamic vs. Static GDOP:

- Static GDOP: For fixed transducer configuration, GDOP varies with target position
- Dynamic GDOP: USV motion changes viewing geometry, improving observability over time

Strategy for SBL Management:

1. Calculate GDOP heatmaps for different modem subsets over operational area
2. Identify regions of poor GDOP where additional modems needed
3. Use USV trajectory planning to minimize time in poor GDOP regions
4. Adaptively switch configurations based on current GDOP and mission requirements

5. Observability Analysis for Underwater Localization

5.1 Observability Fundamentals

Definition and Significance: A system is observable if available measurements uniquely determine all state variables. For underwater positioning, observability determines whether the AUV position is uniquely determinable from available acoustic measurements.

Mathematical Basis: Observability rank condition requires observability matrix O to have full rank:

$$O = \begin{bmatrix} h(x) \\ \nabla h^T(x) \\ \nabla^2 h(x) \\ \vdots \end{bmatrix}$$

Rank deficiency indicates unobservable state directions.

5.2 Observability in Range-Based Acoustic Positioning

Arrichiello et al. Analysis [web:25]: Seminal work on observability metrics for underwater vehicles with range measurements. Key findings:

1. **Single Beacon Limitation:** Single range measurement defines sphere - infinite positions possible
2. **Two Beacons:** Two range measurements define circle - still infinite positions
3. **Three Beacons:** Typically provides unique position in 3D (with depth constraint)
4. **Observability Degradation:** Occurs when relative velocity and position vectors are nearly parallel

Observability Index: Proposed quantitative metric:

$$\frac{\partial d_i}{\partial \theta} \times \cos(\angle(v_{rel}, \vec{r}))$$

where:

- d_i = distance to beacon i
- θ = bearing angle
- v_{rel} = relative velocity
- Smaller angle between velocity and position = worse observability

5.3 Observability in SBL Configuration Subsets

Two-Modem Configuration:

- Cannot achieve full 3D observability in vertical direction
- Useful for 2D horizontal positioning (with depth from pressure sensor)
- Problematic if AUV depth unknown

Three-Modem Configuration:

- Typically achieves full 3D observability (assuming non-coplanar)
- Minimum for 3D positioning without additional constraints
- Coplanar arrangement (all on water surface) degrades observability

Four-Modem Configuration:

- Provides redundancy and improved robustness
- Allows continued operation with one modem failure
- Better conditioning for state estimation

Moving Baseline Advantage: As USV moves relative to AUV, viewing geometry changes. Observability improves over time as system gains diverse viewing angles. Extended observation window can compensate for suboptimal transducer geometry.

5.4 Observability and GDOP Relationship

Complementary Metrics:

- GDOP: Geometric scaling of measurement errors to position errors
- Observability: Whether position is uniquely determinable

Configuration that is observable but has poor GDOP: Position determinable but with large error

Configuration that is unobservable: Position indeterminate regardless of measurement quality

6. Multi-Sensor Fusion: Acoustic, DVL, and IMU Integration

6.1 Sensor Characteristics and Complementary Properties

Acoustic Positioning (SBL):

- Provides absolute position relative to surface baselines
- Update rate limited by modem bandwidth and power (1-10 Hz typical)

- Errors: multipath, sound speed variability, measurement noise
- Advantage: Absolute position reference

Doppler Velocity Logger (DVL):

- Measures velocity vector (3 components) using bottom acoustic reflection
- High update rate (50+ Hz when available)
- No absolute position information, only velocity
- Requires bottom lock (fails in deep water)
- Advantage: Continuous velocity measurement between position updates

Inertial Measurement Unit (IMU):

- Accelerometer: 3 linear acceleration components
- Gyroscope: 3 angular velocity components
- Very high update rate (100+ Hz)
- Provides dead reckoning capability
- Errors accumulate (drift) over time without absolute reference

6.2 Fusion Architectures

Loosely Coupled Fusion: SBL measurements → Position update → IMU/DVL continue dead reckoning until next position

Advantages:

- Computationally simple
- Modular (can add/remove sensors independently)
- Works with partial sensor data

Disadvantages:

- Discards measurement information (only uses final position)
- Cannot exploit cross-correlations between sensors

Tightly Coupled Fusion: All raw measurements (ranges, velocity, acceleration) → Joint state estimator

Advantages:

- Maximizes information usage
- Better handles measurement uncertainties
- Improved accuracy with correlated measurements

Disadvantages:

- Higher computational load
- More complex implementation
- Requires detailed measurement models

6.3 Extended Kalman Filter (EKF) Implementation

EKF Overview: Extends Kalman filtering to nonlinear systems through linearization around current estimate:

Prediction: $\hat{x}_{k|k-1} = f(\hat{x}_{k-1}, u_k)$ $P_{k|k-1} = F_k P_{k-1} F_k^T + Q_k$

Update: $y_k = z_k - h(\hat{x}_{k|k-1})$ $S_k = H_k P_{k|k-1} H_k^T + R_k$ $K_k = P_{k|k-1} H_k^T S_k^{-1}$ $\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k y_k$

Application to AUV Navigation: States: Position (3), velocity (3), orientation (3) = 9-state system (or 6 for 2D + depth)

Process model: Kinematic equations based on thruster commands Measurement models:

- SBL: ranges to each active modem
- DVL: velocity vector
- IMU: acceleration and angular velocity

6.4 Handling Sensor Degradation

DVL Bottom Lock Loss: Levy and Klein [web:12] propose using DVL beam data to estimate acceleration even without bottom lock:

1. Calculate velocity change from Doppler shift alone
2. Differentiate to get acceleration
3. Use acceleration as measurement in EKF

Enables INS/DVL fusion to continue operating during partial DVL outages.

Partial Measurement Integration: Extended Loose Coupling (ELC) approach allows use of incomplete measurement sets. For example, if one acoustic modem fails, use remaining modem measurements as partial constraint on position.

Adaptive Covariance Tuning: During sensor failures or degraded conditions, increase measurement noise variance (R matrix) to reduce influence of unreliable measurements:

$R_{\text{adjusted}} = R_{\text{nominal}} \times (1 + \text{degradation factor})$

6.5 Recent Advances in Multi-Sensor Fusion

Modular Fusion Architecture: Scheiber et al. [web:16] developed MaRS (Modular Underwater Navigation System) allowing flexible sensor combination:

- GNSS (surface) / acoustic modem (underwater)
- Pressure sensor (depth)
- IMU (acceleration/gyration)
- Magnetometer (heading)

Demonstrated improved depth estimation by fusing all sensors compared to individual sensor use.

Unscented Kalman Filter (UKF): Alternative to EKF providing better handling of nonlinearities through sigma point approach:

- UKF typically outperforms EKF for highly nonlinear systems
 - Computational cost scales as $O(n^3)$ vs. EKF $O(n^2)$
 - For small-state underwater vehicles ($n=6-9$), computational difference negligible
-

7. Energy Management and Power Optimization

7.1 USV Battery and Power System Constraints

Lithium-Ion Battery Characteristics:

- Energy density: 150-250 Wh/kg typical
- Round-trip efficiency: 85-95%
- Depth rating: Pressurized housings for deep-water batteries
- Battery Management System (BMS) essential for safe operation

Example Calculation: 10 kg LiPo battery pack $\times$ 200 Wh/kg = 2000 Wh = 2 kWh usable energy

Modem Power Consumption: Typical acoustic modem power budget:

- Transmit (5W) $\times$ 20% duty cycle = 1.0W
- Receive (0.26W) $\times$ 30% duty cycle = 0.078W
- Standby (10mW) $\times$ 50% duty cycle = 0.005W
- Average per modem: $\sim$ 1.08W

For 4 modems continuously active:

- Average power: 4.32W
- Daily consumption: 103.7 kWh (exceeds small USV battery)

Implication: Selective modem activation essential. Duty cycling or selective transducer operation needed for extended missions.

7.2 Smart Energy Management Systems

Framework (Sequeira et al. [web:21]): Comprehensive approach to extend USV mission duration:

1. Photovoltaic Integration:

- Flexible solar panels rated 100W each
- Flexible panels reduce weight while expanding coverage area
- Maximum power point tracking (MPPT) optimizes power extraction

2. Energy-Aware Path Planning:

- Optimize path not just for distance but for energy consumption
- Account for battery voltage drop over mission duration
- Plan rendezvous points for charging if support vessels available

3. Mission Profile Optimization:

- Adjust speed based on energy budget

- Allocate energy to different subsystems based on mission phase
- Enable/disable non-critical systems (communications, sensors) when possible

Results: Extended mission range by up to 50% through integrated approach combining solar panels and intelligent energy management.

7.3 Modem Power Management Strategies

Transmission Power Adaptation:

- Reduce transmit power when target nearby (reduces range but conserves energy)
- Adaptive transmission: Increase power only when needed for long-range positioning
- Estimated energy savings: 30-50% through adaptive power control

Time-Division Multiplexing:

- Sequential activation of modems instead of simultaneous
- Reduces acoustic interference (improves signal quality)
- Enables smaller transmit power for same SNR
- Trade-off: Slower position update rate

Duty Cycling:

- Periodically deactivate modems during non-critical mission phases
- Resume high-frequency updates during critical operations
- Example: Survey phase (slow movement) → Low-rate positioning; docking phase (fast positioning needed) → High-rate

Selective Transducer Activation:

- Use GDOP analysis to determine minimum sufficient modem set
- Activate additional modems only when current configuration insufficient
- Real-time switching based on achievable GDOP and mission requirements

7.4 Energy-Accuracy Trade-off Optimization

Multi-Objective Optimization: Objective: Maximize mission duration while maintaining positioning accuracy

$$\min_{\{c(t), P(t)\}} \left[E_{\text{consumed}}(c, P) - \lambda \cdot A_{\text{achieved}}(c, P) + \mu \cdot G_{\text{achieved}}(c, P) \right]$$

where:

- $c(t)$ = modem configuration (which active at time t)
- $P(t)$ = transmit power
- E_{consumed} = energy used
- A_{achieved} = positioning accuracy
- G_{achieved} = GDOP metric
- λ, μ = weighting parameters (user-defined)

8. Acoustic Communication Challenges and Limitations

8.1 Multipath Propagation

Physical Cause: Sound waves reflecting off water surface and seafloor create multiple arrival paths. In shallow water (50-500m depth), multipath delay spread can reach 10-100ms.

Effect on Positioning:

- Ambiguous time-of-arrival measurement (which reflection is direct path?)
- Intersymbol interference in digital communications
- Reduced effective SNR of acoustic signal

Mitigation Techniques:

1. **Equalization:** Channel estimation and decision-feedback equalization
2. **Error Correction:** Coding to recover from multipath-induced errors
3. **Frequency Diversity:** Spread signal over wide bandwidth to reduce multipath coherence
4. **Adaptive Equalization:** Adjust receiver filters based on observed channel
5. **Pulse Shaping:** Design transmitted pulse to minimize multipath sensitivity

8.2 Sound Speed Variability

Environmental Dependence:

- Freshwater: ~1480 m/s
- Seawater: ~1500 m/s
- Variation with temperature: -2 to +4 m/s per °C
- Variation with salinity: +1.3 m/s per ppt

Impact on Ranging: Range error = $\Delta C \times \tau / 2$, where τ is measured time-of-flight

Example: 0.5% sound speed error (7-8 m/s) over 1000m range

- Measured time: ~1.33 seconds
- Measured range error: ± 3.5 meters

Solutions:

1. **Direct Measurement:** CTD (Conductivity-Temperature-Depth) sensor measures sound speed
2. **Lookup Tables:** Use temperature/salinity profile to estimate sound speed
3. **Estimation in Filter:** Include sound speed as state variable in Kalman filter
4. **Two-Path Transmission:** Use different frequencies to bracket sound speed range

8.3 Bandwidth and Data Rate Limitations

Commercial Modem Specifications:

- Low frequency (15-27 kHz): 6-9.2 kbit/s, range 6000m
- Mid frequency (42-65 kHz): 31.2 kbit/s, range 1000m
- High frequency (120-180 kHz): 62.5 kbit/s, range 300m

Bandwidth Constraint: Underwater acoustic channel bandwidth limited by multipath characteristics and absorption:

$$B_{\max} \approx \frac{c}{\tau_{\text{spread}}}$$

where τ_{spread} is multipath delay spread. In shallow water: $B_{\max} \sim 10\text{-}100$ kHz.

Impact on SBL:

- Position update rate limited by modem bandwidth
- Typical SBL position rate: 1-10 Hz (period 0.1-1 second)
- Slower than IMU/DVL update rates, requiring fusion for continuous estimates

Mitigation:

1. **Efficient Coding:** Use variable-rate codes to maximize information per transmission
2. **Compressed Measurements:** Transmit only measurement residuals
3. **Predictive Transmission:** Modem predicts position change, transmits only correction

8.4 Noise and Interference Sources

Ambient Noise:

- Biological: Snapping shrimp noise (dominant in shallow warm water, 2-20 kHz)
- Shipping: Low-frequency noise from large vessel propellers
- Wind/Wave: Surface noise

Equipment Noise:

- Propeller cavitation (onboard engines)
- Electronic noise from power systems
- Cross-talk between multiple modems

SNR Impact: Signal-to-Noise Ratio directly affects ranging accuracy through Cramer-Rao bound:

$$\text{CRLB}_{\text{range}} \approx \frac{c^2}{2\pi f_c \sqrt{\text{SNR} \cdot T}}$$

where f_c is center frequency, T is observation duration.

9. Array Design and Configuration Optimization

9.1 Theoretical Principles of Array Optimization

Array Signal Processing Fundamentals: Acoustic arrays process signals from multiple transducers to enhance signal detection and directionality. Key metrics:

1. **Beamwidth:** Angular width of main lobe (narrower = better resolution)
2. **Sidelobe Level:** Energy in secondary lobes (lower = better interference rejection)
3. **Directivity:** Measure of angular selectivity
4. **Gain:** Signal amplification through coherent combination

Vector Sensor Arrays (Kitchens [web:79]): Advanced approach measuring both acoustic pressure (scalar) and particle velocity (vector). Advantages:

- Resolves front/back ambiguity inherent in pressure sensors
- Better performance in interference-limited scenarios
- More compact than pressure-only arrays of equivalent capability

Challenges:

- Vector sensors more sensitive to orientation errors
- Requires careful calibration
- Processing algorithms more complex

9.2 Modem Array Configuration for SBL

Spatial Configuration Options:

1. Linear Array:

- Modems arranged in line
- Good resolution along array axis
- Poor resolution perpendicular to array
- Singular configurations possible (target on line through array)

2. Triangular/Tetrahedral Configuration:

- Modems at vertices of triangle (2D) or tetrahedron (3D)
- More uniform performance in different directions
- Better GDOP characteristics
- Physically feasible on USV (2D possible, 3D difficult)

3. Distributed Configuration:

- Modems at different positions on USV (bow, stern, sides)
- Leverages USV geometry for modem separation
- Best practical implementation for hull-mounted systems

9.3 Parametric Acoustic Arrays

Novel Concept (Jiang et al. [web:82]): Parametric acoustic arrays achieve high directionality with small apertures through nonlinear acoustics:

- Primary high-frequency carriers (30-100 kHz)
- Intermodulation products generate lower frequencies
- Effective aperture much larger than physical size

Application to SBL:

- Could enable single compact element providing multi-transducer equivalent
- Reduces power consumption (single transmitter vs. multiple)
- Enables tunable directivity (adjusting carrier frequencies)

Current Limitations:

- Technology still developing
- Efficiency concerns for modem applications
- Complexity vs. benefit trade-off not yet resolved

9.4 Time-Varying Array Configuration

Dynamic Reconfiguration: Rather than fixed modem subset, dynamically adjust which modems are active:

Decision Logic:

1. Calculate current GDOP for all active modems
2. If GDOP exceeds threshold, activate additional modems
3. If GDOP acceptable and power critical, deactivate redundant modems
4. Use hysteresis to prevent rapid switching

Challenges:

- Acoustic interference if modems transmit simultaneously
 - Time-division multiplexing reduces effective bandwidth
 - State estimation disruption during configuration transition
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10. Observability-Based Adaptive Control

10.1 Observability as Decision Criterion

Beyond Accuracy Metrics: While GDOP predicts accuracy, observability determines whether position is uniquely determinable at all. Adaptive strategy should prioritize observability.

Decision Hierarchy:

1. **Primary:** Is system observable? (Can we determine position uniquely?)
2. **Secondary:** Is GDOP acceptable? (Will accuracy meet mission requirements?)
3. **Tertiary:** Energy efficiency (minimize power while satisfying 1 and 2)

Implementation:

- Calculate observability rank for current modem configuration
- If rank deficient, add modems until observable
- Subject to that constraint, optimize for GDOP and energy efficiency

10.2 Reconfiguration Strategy

Continuous Monitoring: Real-time calculation of:

- Current GDOP (geometry-dependent)
- Observability rank (configuration-dependent)
- Mission accuracy requirement
- Available battery capacity

Decision Algorithm:

```

if observability_rank < 3:
    activate additional modems until rank = 3
else:
    if GDOP > GDOP_threshold:
        activate additional modems
    else:
        if battery_low:
            deactivate non-critical modems
        end if
    end if
end if

```

Hysteresis to Prevent Chattering:

- Activation threshold ($GDOP > 10$): Activate modem 4
- Deactivation threshold ($GDOP < 5$): Deactivate modem 4
- Prevents rapid oscillation in configuration

10.3 Safe Transitions During Reconfiguration

Challenge: During transition from 2-modem to 3-modem configuration, system temporarily has partially redundant measurements.

Solutions:

1. **Smooth Transition:** Gradually increase weight on newly activated modem
2. **Residual Checking:** Monitor measurement residuals during transition
3. **Covariance Reset:** Allow filter covariance to reset after transition
4. **Validation Gate:** Reject measurements outside expected range during transition

11. USV-AUV Collaborative Operations**11.1 System Architecture for Collaborative Missions****Typical Configuration:**

- USV on surface: Carries acoustic modems, provides communication relay
- AUV underwater: Carries transponder, performs mission tasks
- Communication: Via acoustic modem (typically 250-4500 bit/s)
- Coordination: Pre-programmed rendezvous, waypoints, or real-time commands

Joint Mission Examples:**1. Survey Operations:**

- AUV performs detailed mapping/sampling at survey site
- USV maintains position overhead, provides positioning

- Relative positioning between units typically 100-1000m

2. Inspection/Maintenance:

- AUV performs underwater inspection/repair
- USV provides support: power, tools, positioning
- Closer cooperation, relative positions 10-100m

3. Collaborative Sensing:

- Multiple AUVs form swarm
- USV provides acoustic navigation support for entire swarm
- SBL used to maintain formation geometry

11.2 Communication Bandwidth Constraints

Data Rate Limitation: At 250 bit/s modem rate:

- Simple position message (8 bytes): 256 bits → 1 second transmission
- Velocity/status (4 bytes): 128 bits → 0.5 second transmission
- Control command (2 bytes): 16 bits → 60ms transmission

Implication for Control: Real-time closed-loop control with direct acoustic link impractical. Operations typically:

- Pre-planned with waypoints
- Execute autonomously with periodic status updates
- Manual control only during critical maneuvers

11.3 Multi-AUV Formation Control

Formation Geometry Maintenance: Multiple AUVs must maintain formation shape for coordinated operations. Uses:

- Range-only formation control (measured distances between units)
- Acoustic modem-based ranging between AUVs
- Extended Kalman filter for relative position estimation

Energy Implications: Additional ranging measurements between AUVs consume bandwidth and power. Strategy:

- Use relative positioning during formation maintenance
- Switch to SBL from USV during formation acquisition
- Trade off energy for formation accuracy based on mission phase

11.4 Real-World Demonstrations

DriX/Ulyx Mission (Exail/Ifremer [web:71]): First successful joint USV-AUV deep-water mission:

- USV (DriX) surface platform
- AUV (Ulyx) capable of 6000m depth
- Acoustic communication enabled real-time coordination

- Demonstrated viability of collaborative deep-sea operations

Key Success Factors:

1. Modular acoustic modem integration (WHOI Micromodem)
 2. Autonomous operation with periodic updates
 3. Backup procedures for communication loss
 4. Pre-mission planning and validation
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12. Mission Planning and Optimization

12.1 Path Planning Accounting for Energy Constraints

Energy-Optimal Path Planning: Traditional path planning minimizes distance. Energy-optimal planning must account for:

- Current and wind (variable by location)
- Modem power consumption patterns
- Battery voltage effects on system efficiency

Level Set Methods (Subramani [web:81]): Fast marching and level set methods solve time/energy-optimal path planning by:

1. Creating cost map based on energy requirement at each point
2. Computing shortest path in cost metric
3. Scalable solution (linear in number of vehicles)

Application to SBL Management: Plan USV trajectory to:

1. Maintain observable geometry for AUV positioning
2. Minimize time spent in high-GDOP regions
3. Account for energy consumed by positioning system
4. Enable more efficient mission overall

12.2 Cooperative Mission Planning

Multi-Agent Coordination: Research on heterogeneous systems (UAV-USV-AUV) demonstrates approaches for:

- Task allocation among agents
- Resource sharing (battery charging, communication bandwidth)
- Failure recovery and re-planning

Example: Multi-AUV Survey (Li et al. [web:69]):

- Three AUVs executing survey missions
- Two USVs serving as mobile battery chargers
- Optimization determines:
 - Which AUV surveys which area
 - Where rendezvous points located

- When charging occurs

Results:

- 205.3km total AUV travel distance covered
- Mission completed in 91.4 hours with 8-hour charging
- Reduced to 31 hours average with 3-hour charging

Energy Considerations:

- Longer survey legs require more battery capacity
- Frequent charging interrupts mission but preserves flexibility
- Optimization balances mission duration and flexibility

12.3 Adaptive Mission Re-Planning

Real-Time Adaptation: Initial plan may not account for:

- Actual environmental conditions (stronger currents than predicted)
- Sensor performance degradation
- Hardware failures

Adaptive Replanning Trigger: If deviation from plan exceeds threshold (e.g., 10% slower progress), replan:

1. Recalculate energy budget based on actual performance
2. Adjust remaining waypoints
3. Modify charging schedule if multi-AUV mission

SBL Impact: Positioning accuracy affects planning confidence:

- High GDOP/poor accuracy → increase safety margins, more frequent navigation updates
- Good GDOP/high accuracy → can plan tighter trajectories, fewer updates

13. State-of-the-Art Analysis and Research Gaps

13.1 Current Capabilities

Mature Technologies:

- SBL positioning systems with 1-3m accuracy
- Extended Kalman Filter fusion for multi-sensor integration
- GDOP analysis for geometry assessment
- Modem technology with data rates up to 60+ kbit/s
- USV-AUV operational demonstrations

Practical Deployments:

- Offshore oil/gas industry uses SBL for subsea positioning
- Research institutions conduct routine AUV surveys with acoustic navigation
- Military unmanned systems operational with acoustic communication

13.2 Current Limitations and Challenges

Technical Gaps:

1. Limited Modem Options:

- Few modems optimized for low-power SBL operation
- Most designed for communication rather than positioning
- Trade-off between data rate and power not fully explored

2. Incomplete Fusion Algorithms:

- Standard EKF assumes Gaussian noise (underwater: non-Gaussian in presence of multipath)
- Few implementations with adaptive configuration switching
- Limited work on observability-aware filtering

3. Acoustic Environment Adaptation:

- Sound speed variation not always addressed
- Multipath typically mitigated rather than exploited
- Limited work on learning acoustic channel characteristics

4. Energy Management Integration:

- Most research treats energy as afterthought
- Few implementations with real-time energy-aware modem switching
- Battery model often simplified

13.3 Open Research Questions

Fundamental Questions:

1. Optimal Configuration Switching Strategy:

- What decision threshold (GDOP value) triggers configuration change?
- How do hysteresis parameters affect mission energy/accuracy trade-off?
- Can machine learning improve decision-making based on mission data?

2. Observability-Aware State Estimation:

- How to incorporate observability constraints into filter design?
- Can observability metric directly guide measurement selection?
- What filter architecture best exploits partial observability?

3. Multipath Exploitation:

- Can multipath reflections be used productively for positioning?
- What receiver architectures enable multipath exploitation?
- What is energy cost of multipath-aware processing?

4. Predictive Energy Models:

- How to predict actual modem power consumption in specific environments?

- Can environmental measurements (salinity, temperature) predict energy needs?
 - What is optimal energy allocation across mission phases?
-

14. Synthesis and Framework Development

14.1 Integrated Framework Components

Proposed Adaptive SBL Management Framework:

1. Real-Time Perception:

- Current modem configuration (2, 3, or 4 active)
- Latest FIM analysis results
- GDOP metric for current geometry
- Battery state and projected remaining capacity
- Current and predicted AUV position/velocity

2. Decision Logic:

- Observability constraint satisfaction (minimum configuration)
- GDOP-based configuration adjustment
- Mission accuracy requirements
- Energy budget constraints
- Hysteresis-based switching to prevent chattering

3. Execution:

- Activate/deactivate modems according to decision
- Adjust transmit power if needed
- Replan USV trajectory to improve GDOP if beneficial
- Update state estimator with new configuration

4. Learning/Adaptation:

- Log actual performance vs. predicted
- Update noise models based on observations
- Adjust decision thresholds based on mission outcomes

14.2 Implementation Roadmap

Phase 1: Simulation Validation (6 months)

- Develop simulator with realistic acoustic channel, modem models
- Implement decision algorithms
- Validate against benchmark mission profiles
- Compare adaptive vs. fixed configurations

Phase 2: Laboratory Testing (6 months)

- Implement algorithms on AUV/USV test platforms
- Conduct tank trials with controlled geometry

- Validate real-time performance
- Measure actual power consumption vs. models

Phase 3: Sea Trials (12 months)

- Deploy in shallow water first (better acoustic conditions)
- Validate in realistic noisy environment
- Extend to deeper water operations
- Document lessons learned and algorithm refinements

Phase 4: Operational Deployment (ongoing)

- Integration into operational AUV systems
- Continuous improvement based on mission data
- Collaboration with industry partners
- Technology transition to commercial modem manufacturers

14.3 Performance Metrics and Success Criteria

Energy Efficiency:

- Energy consumed per position update (Joules/update)
- Mission duration extension vs. fixed configuration (%)
- Power savings from selective modem activation (%)

Positioning Accuracy:

- Mean positioning error vs. mission requirement compliance
- GDOP achieved vs. theoretical minimum
- Observability maintained (rank = expected)

System Robustness:

- Performance with single modem failure
- Adaptation time to configuration change
- Long-term drift in position estimates

Operational Impact:

- Ease of parameter tuning for different missions
- Compatibility with existing AUV/USV platforms
- Training required for operators

15. Conclusion

The development of an adaptive SBL management framework addresses a critical practical need: how to maintain accurate underwater vehicle positioning while respecting battery energy constraints on unmanned surface vehicles. The framework synthesizes research across multiple domains:

1. **Acoustic Positioning Theory:** FIM and GDOP analysis provide quantitative tools for configuration assessment

2. **State Estimation:** Multi-sensor fusion with Kalman filtering enables robust position estimates
3. **System Observability:** Ensures fundamental possibility of position determination
4. **Energy Management:** Intelligent modem duty-cycling extends mission duration
5. **Collaborative Operations:** Supports efficient USV-AUV teamwork

The research is timely, addressing growing needs for extended autonomous operations in oceanographic research, subsea infrastructure monitoring, and maritime defense applications. By intelligently managing acoustic modem configurations, significant improvements in both positioning accuracy and energy efficiency are achievable, enabling longer-duration, more capable unmanned operations.

Key Research References

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This comprehensive literature review provides the scientific foundation for developing adaptive short-baseline acoustic positioning management systems. The integrated framework combining FIM analysis, observability assessment, multi-sensor fusion, and energy-aware control offers a path toward more capable and efficient unmanned underwater operations.